Routing in React (React Router)

Question 1: What is React Router? How does it handle routing in single-page applications?

Answer: React Router is the standard routing library for React. In a Single-Page Application (SPA), there are no actual physical HTML pages (about.html / contact.html) to load from a server. Instead, React Router intercepts the browser's URL changes. When the URL changes, it dynamically hides the current component and displays a new component matching that path—all instantly, without triggering a full page refresh.

Question 2: Explain the difference between BrowserRouter, Route, Link, and Routes (formerly Switch) components in React Router.

Answer:

- **BrowserRouter**: The parent wrapper component that enables routing across your entire application. It keeps your UI in sync with the browser's address bar URL using the HTML5 history API.
- **Routes (formerly Switch)**: The container component that looks through all its child <Route> elements and renders **only the first one** that matches the current URL path.
- **Route**: The component that maps a specific URL path to a specific component. It takes a path prop (e.g., /about) and an element prop (the component to show).
- **Link**: The component used to navigate between views. It renders an HTML <a> tag behind the scenes but prevents the browser from reloading the page, keeping the application state intact.

LAB EXERCISE

Task 1: Set up a basic React Router with routes for a Home page and an About page. Display the appropriate content based on the URL.

```
import React from 'react';
import { BrowserRouter, Routes, Route } from 'react-router-dom';

// 1. Create simple page components
function Home() {
  return <h2>Welcome to the Home Page!</h2>;
}

function About() {
  return <h2>Learn more About Us on this page.</h2>;
}

// 2. Set up the router structure
function AppRouter() {
  return (
    <BrowserRouter>
      <div style={{ padding: '20px', fontFamily: 'sans-serif' }}>
        {/* The Routes container ensures only one view renders at a time */}
        <Routes>
          <Route path="/" element={<Home />} />
          <Route path="/about" element={<About />} />
```

```
        </Routes>
      </div>
    </BrowserRouter>
  );
}

export default AppRouter;
```

Task 2: Create a navigation bar using React Router's Link component that allows users to switch between the Home, About, and Contact pages.

```
import React from 'react';
import { BrowserRouter, Routes, Route, Link } from 'react-router-dom';

// Simple component views
const Home = () => <h2>Home View</h2>;
const About = () => <h2>About View</h2>;
const Contact = () => <h2>Contact View</h2>;

function App() {
  return (
    <BrowserRouter>
      <div style={{ fontFamily: 'sans-serif', padding: '20px' }}>

        {/* Navigation Bar Header */}
        <nav style={{
          backgroundColor: '#333',
          padding: '10px',
          borderRadius: '5px',
          marginBottom: '20px'
        }}>
          {/* Using Link instead of <a href> avoids page reloads */}
          <Link to="/" style={linkStyle}>Home</Link> |
          <Link to="/about" style={linkStyle}>About</Link> |
          <Link to="/contact" style={linkStyle}>Contact</Link>
        </nav>

        {/* Dynamic Route Display */}
        <Routes>
          <Route path="/" element={<Home />} />
          <Route path="/about" element={<About />} />
          <Route path="/contact" element={<Contact />} />
        </Routes>

      </div>
    </BrowserRouter>
  );
}

// Simple layout style object for navigation links
const linkStyle = {
  color: 'white',
  textDecoration: 'none',
  margin: '0 15px',
```

```
  fontWeight: 'bold'
};

export default App;
```